



Heaven's Light is Our Guide

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF MECHATRONICS ENGINEERING

Course No.: **MTE 4200**

Course Title: **Project and Thesis**

Tactile and Proprioceptive Shape Modeling For Planning Visionless Grasp With Reinforcement Learning In A Soft Robotic Manipulator

PRESENTED BY

ABRAR FAIYAZ ANAN CHOWDHURY

ID: **2008008**

AVIK MD. EMTIAZ AREFIN

ID: **2008013**

SUPERVISED BY

MD ZUNAID HOSSEN

Lecturer

Department of Mechatronics Engineering
Rajshahi University of Engineering & Technology



OUTLINE

- Introduction
- Problem
- Objectives
- Literature Review
- Solution
- Methodology
- Schematic Diagram
- Mathematical Modeling
- Project Development
- Experimental Setup
- Result
- Validation
- Analysis
- Impact
- Project Management
- Discussion & Limitations
- Future Work
- Conclusion

INTRODUCTION

Transitioning from
vision dependent to **vision
constrained** manipulation

Intelligent Touch

“Octopuses have impressive visionless manipulation capabilities by the intrinsic adaptation of their soft and variable stiffness arms, combined with distributed suckers”

THE PROBLEM

Visual Failure
in Darkness

Lack of Optimal Grasp
in Vision Constrained Environments

Instability and Damage
in Visionless Rigid Robots



OBJECTIVES

1

To develop a visionless shape modeling framework to fuse sequential tactile and proprioceptive sensing data for achieving **a coarse 3D reconstruction** with high accuracy.

2

To optimize an object specific visionless grasp policy via Reinforcement Learning that **maximizes optimal grasp stability** while penalizing slippage to ensure non damaging manipulation.



LITERATURE REVIEW

REF.	PAPER NAME	JOURNAL	YEAR	METHODOLOGY	FINDINGS	RESEARCH GAP
[1]	Embodied Intelligence Technology for Next Generation Soft Robots	<i>Engineered Science</i>	2026	• Co Design Paradigms • Perception Control Triad Framework	• Maps integration across soft materials, sensing and RL controllers	• Lack of unified real time soft body simulation frameworks
[2]	Proprioceptive State Estimation for Amphibious Tactile Sensing	<i>IEEE T-RO</i>	2025	• In Finger Camera Tracking • Volumetric Discretized Soft Body Model	• Sub 2 mm accurate deformation tracking on land, underwater	• Computational overhead of high framerate image processing

[1] J. Zhang, R. Wang, and Y. Sun, "Embodied intelligence technology for next generation soft robots," *Engineered Science*, vol. 40, pp. 2073–2088, 2026

[2] X. Han, N. Guo, S. Zhong, Z. Zhou, J. Lin, C. Song, and F. Wan, "Proprioceptive state estimation for amphibious tactile sensing," *IEEE Transactions on Robotics*, 2025.



LITERATURE REVIEW

REF.	PAPER NAME	JOURNAL	YEAR	METHODOLOGY	FINDINGS	RESEARCH GAP
[3]	Embedding High Resolution Touch Across Robotic Hands Enables Adaptive Human like Grasping	<i>Nature MI</i>	2025	• Photometric Stereo Tactile Skin • Physics based Image Formation Model	• Outperforms vision only hands in complex multi object grasping	• Spatial routing of dense wiring • Sensor durability over long term dynamic wear
[4]	Triaxial Tactile Sensing for Next Gen Robotics	<i>Sci. Explor. Press</i>	2025	• Flexible Dual Inductive Sensing • Real Time Force Decoupling	• Decouples normal and shear forces with 21 ms fast response	• Material degradation over continuous cyclic shear loads

[3] Y. Zhang, X. Liu, and H. Wang, "Embedding high resolution touch across robotic hands enables adaptive human like grasping," *Nature Machine Intelligence*, vol. 7, no. 6, pp. 889–900, 2025.

[4] G. Yun, H. Li, and X. Zhang, "Triaxial tactile sensing for next gen robotics," *Science Exploration Press*, vol. 17, no. 1, pp. 12–25, 2025.



LITERATURE REVIEW

REF.	PAPER NAME	JOURNAL	YEAR	METHODOLOGY	FINDINGS	RESEARCH GAP
[5]	DexGrip: Multi modal Soft Gripper with Dexterous Grasping	IEEE RoboSoft	2025	• Variable Stiffness Actuation • Multi Modal Sensor Fusion	• High payload adaptability with in hand rotation dexterity	• Complex multi channel pneumatic distribution
[6]	A Tendon Driven Continuum Manipulator with Shape Estimation	IEEE RAL	2024	• Distributed IMU Sensor Fusion • b) Kinematic Curvature Constraint	• Drift free shape reconstruction during continuous bending	• Cable friction non linearities under heavy payload

[5] H. Kim, J. Park, and S. Lee, "Dexgrip: Multi modal soft gripper with dexterous grasping," in *IEEE International Conference on Soft Robotics (RoboSoft)*, pp. 102–109, 2025

[6] J. Lu, Y. Chen, and Z. Wang, "A tendon driven continuum manipulator with robust shape estimation," *IEEE Robotics and Automation Letters*, vol. 9, no. 2, pp. 1420–1427, 2024.

LITERATURE REVIEW

Table 1: Comparison Table between Top 6 Papers & Proposed Thesis

FEATURES	[1]	[2]	[3]	[4]	[5]	[6]	[PROPOSED THESIS]
<i>Visionless Grasping</i>	✗	✓	✓	✗	✓	✗	✓
<i>Tactile Sensing</i>	✓	✗	✗	✓	✓	✓	✓
<i>Proprioceptive Sensing</i>	✗	✓	✓	✗	✓	✗	✓
<i>Reinforcement Learning</i>	✓	✗	✓	✗	✗	✗	✓
<i>Adaptive Grasp Control</i>	✗	✗	✓	✗	✓	✗	✓
<i>Soft Robotics Application</i>	✓	✗	✗	✓	✗	✗	✓
<i>Shape Reconstruction</i>	✗	✓	✗	✗	✓	✓	✓
<i>Biological Inspiration</i>	✓	✗	✓	✓	✗	✗	✓

SOLUTION

D.

INTRODUCING

Dark_bot

A Visionless Soft Robotic Manipulator

Visionless

Flexible

**Coarse 3D
Geometry**

**Optimal
Grasp**



METHODOLOGY

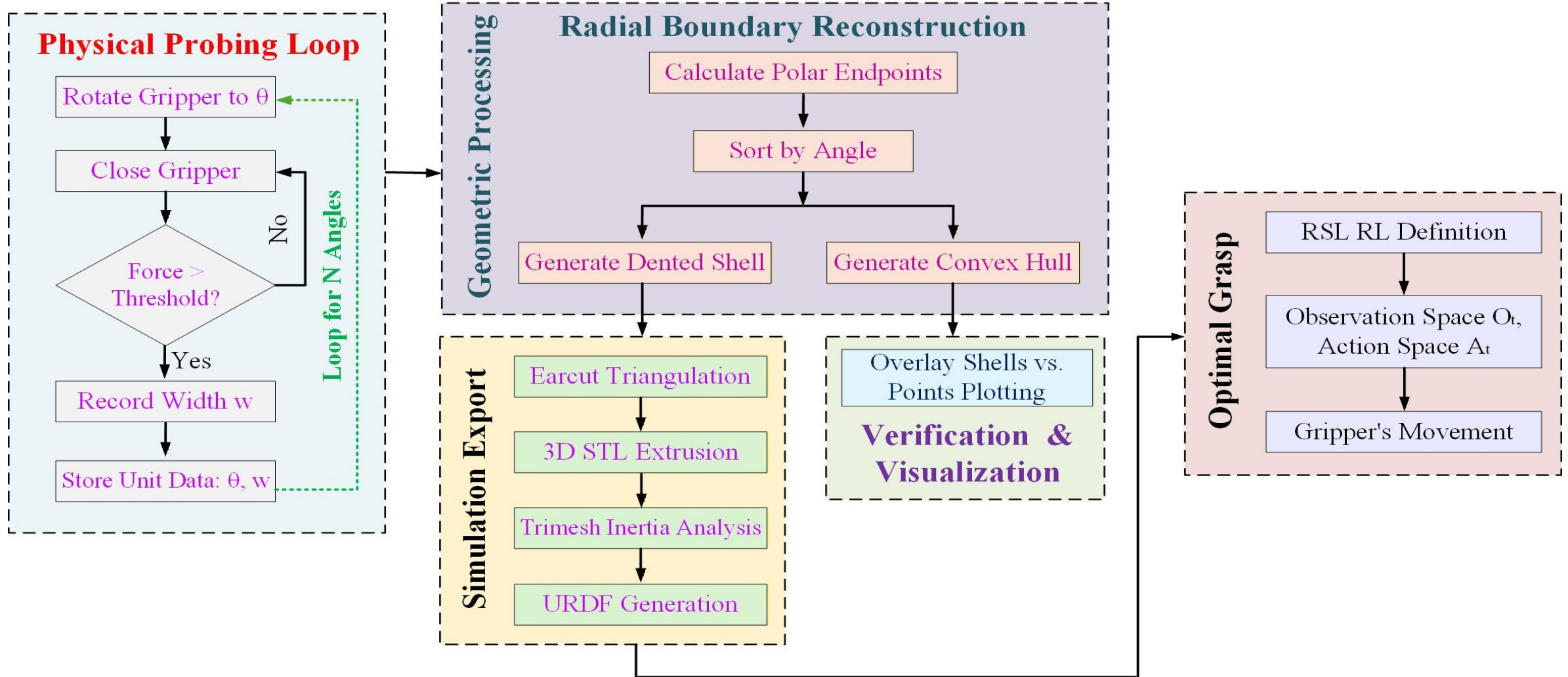


Figure 1: Step by Step Workflow of the Proposed Approach

COMPONENTS



Figure 2: Hiwonder 6DOF Metal Robotic Arm with Controller



Figure 3: Raspberry Pi 5 Model 16GB



Figure 4: Flex Sensors 2.2" Original x6



Figure 8: Force Sensing Resistor Flexible Thin Pressure Sensor MD30 60 150g to 20kg 30mm x6



Figure 9: Silicone Rubber Compound with Hardner



Figure 5: Analog Feedback Servo MG996R x5



Figure 6: 16 Channel 12 bit PWM/Servo Driver I2C Interface PCA9685 x1

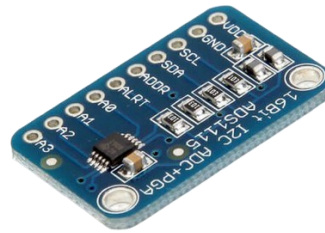


Figure 7: ADS1115 ADS 1115 16 Bit I2C 4 Channel ADC Module x1

SCHEMATIC DIAGRAM

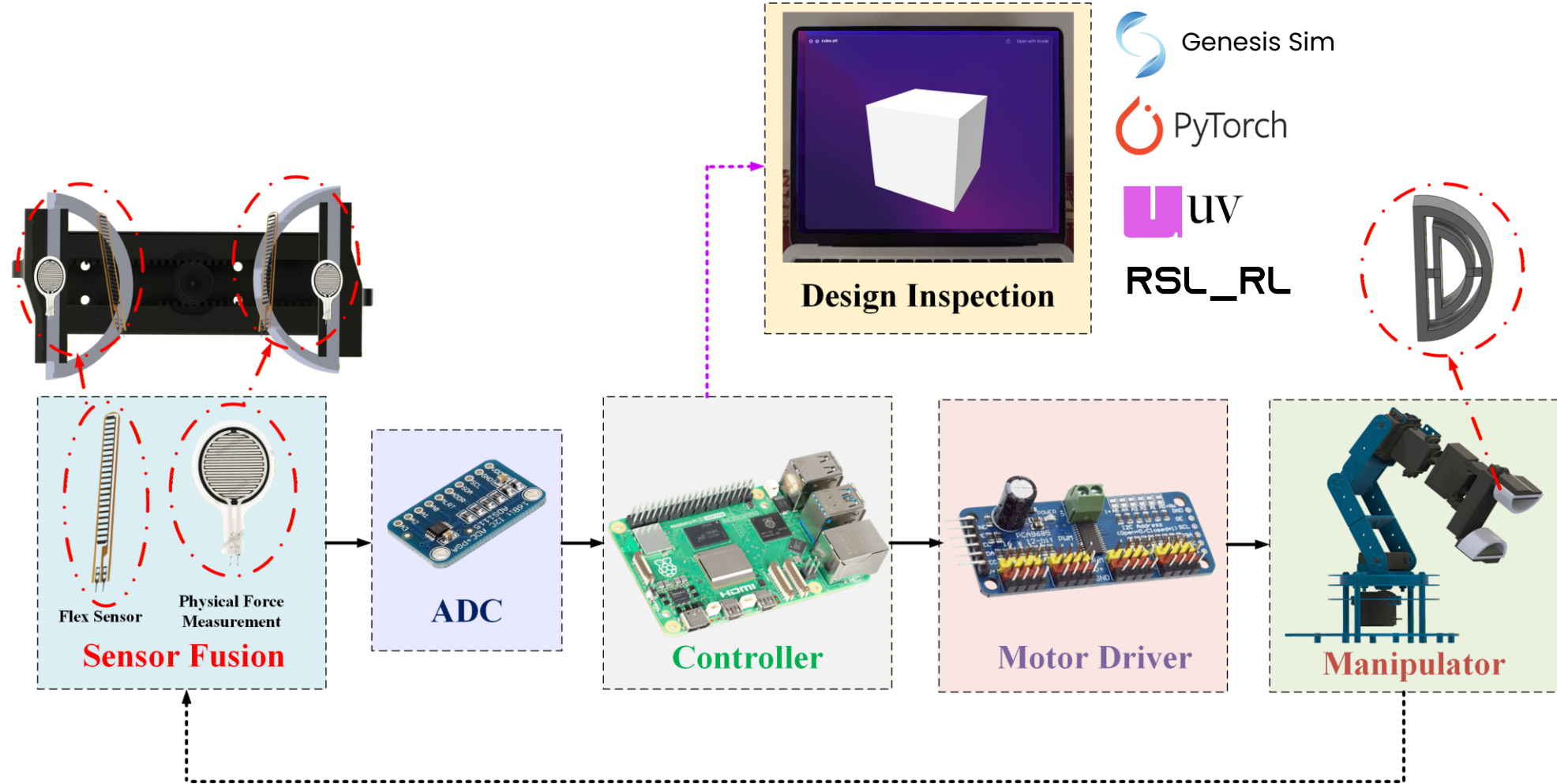


Figure 10: Detailed Hardware Interconnection and Schematic Diagram of the Soft Robotic Manipulator System

MATHEMATICAL MODELING

A. Radial Vertex Generation

A set of triplets $S = \{(\theta_i, w_i, \sigma_i)\}_{i=1}^n$

r , Radial Distance:

$$r_i = \frac{w_i}{2} + \delta_{soft}$$

Left Projection:

$$V \leftarrow (r_i \cos \theta_i, r_i \sin \theta_i)$$

Right Projection:

$$V \leftarrow (-r_i \cos \theta_i, -r_i \sin \theta_i)$$

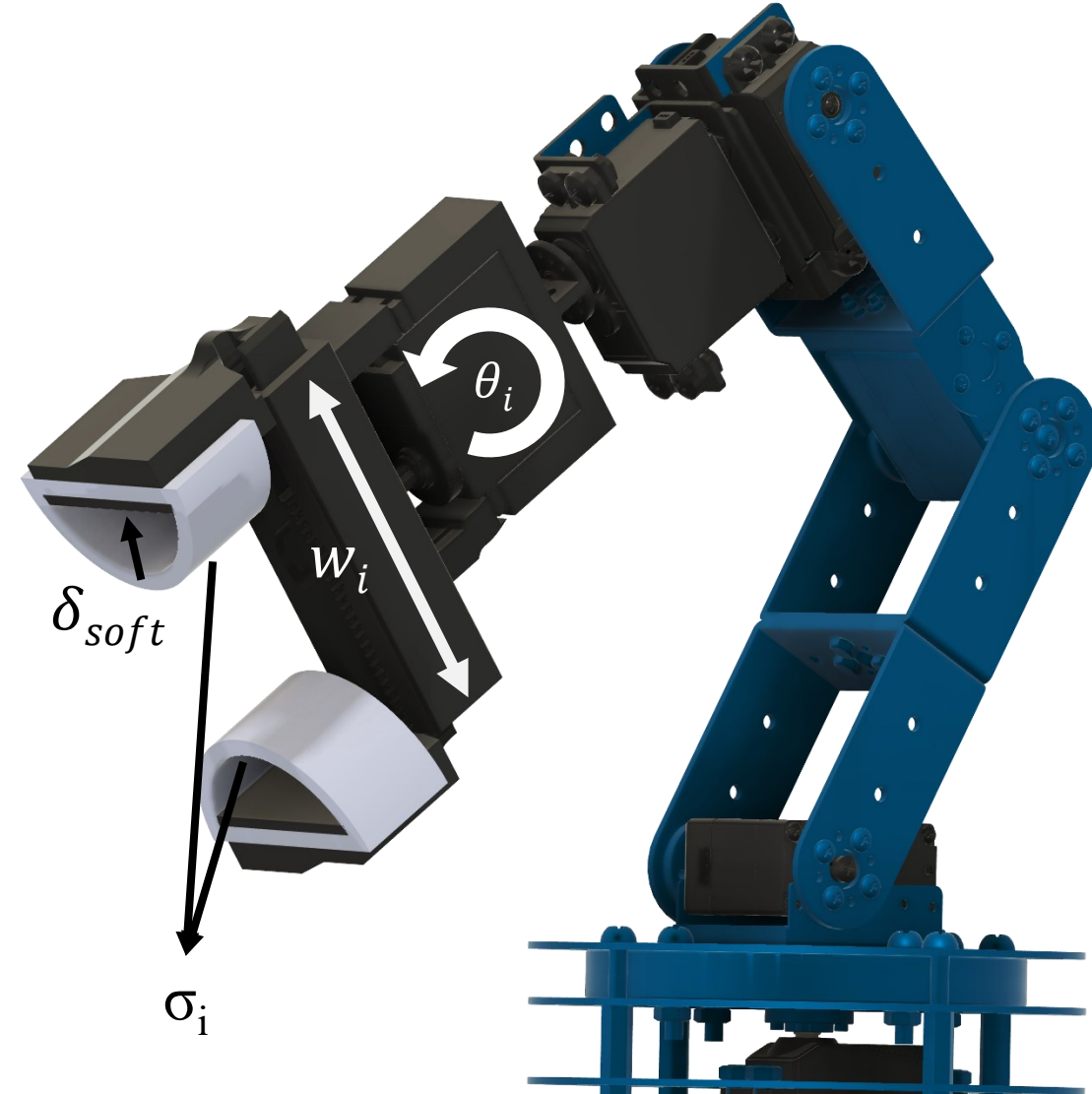


Figure 11: 2D System Modeling



MATHEMATICAL MODELING

B. Topological Ordering and Dented Profile

$$V_{sorted} \leftarrow \text{sort}(V, \text{by } \text{atan2}(y, x))$$

C. Volumetric Dynamics and URDF Export

$$V_{vol} = \frac{1}{6} \left| \sum_{f \in \mathcal{F}} (v_1 \cdot (v_2 \times v_3)) \right|$$

Given a uniform density ρ ,

$$M = \rho V_{vol}$$

The inertia tensor I is calculated. These parameters are serialized into a **URDF file** $\mathcal{U}$, which includes collision and visual tags.



MATHEMATICAL MODELING

D. RL Execution Loop Formulation

Proprioceptive and Tactile Observation Vector,

$$\mathbf{o}_t = \left[P_{2D}, z_{obj}, \theta_{wrist}, w_{grip} \right]^T$$

E. Action Space Constraints and Squeeze Primitive

The environment policy writes **$[-1, 1]$** squeeze value from RL, which translates to **$[0^\circ, 245^\circ]$**

$$\mathcal{A} \in \mathbb{R}^2$$



MATHEMATICAL MODELING

F. Reward Evaluation

Success Indicator	Alignment Reward	Periphery Width Reward
$S = \mathbb{I}(z_{obj} > 0.1)$	$r_{align} = e^{-5.0 \cdot \Delta\theta_{min}}$	$r_{width} = e^{-20.0 \cdot w_{periphery}}$

$$\mathcal{R}(o_t, a_t) = (r_{align} \cdot r_{width}) + (S \cdot r_{align})$$

HARDWARE DEVELOPMENT

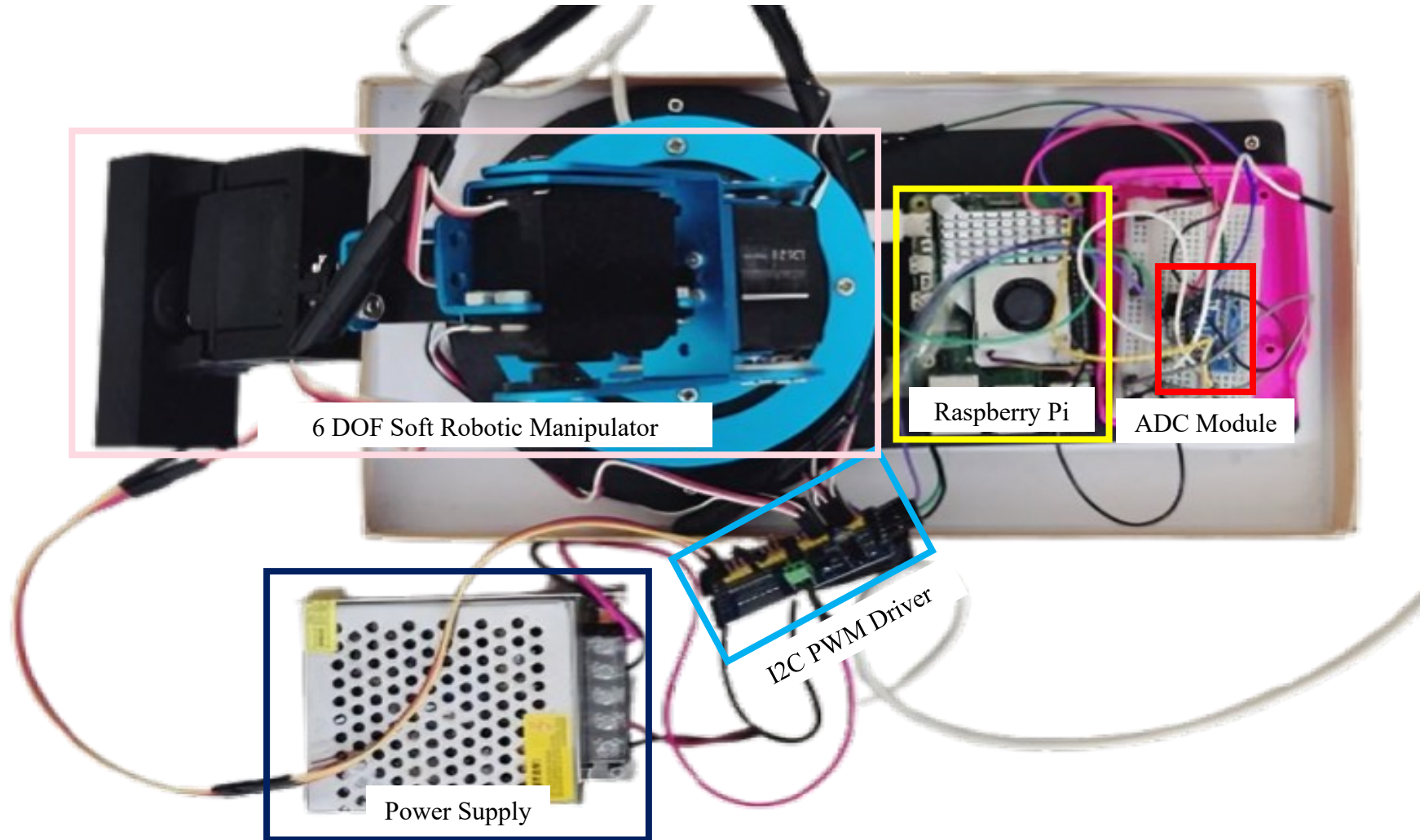


Figure 12: Physical Layout of the Visionless Soft Robotic Manipulator System

SOFT GRIPPER FINGERS

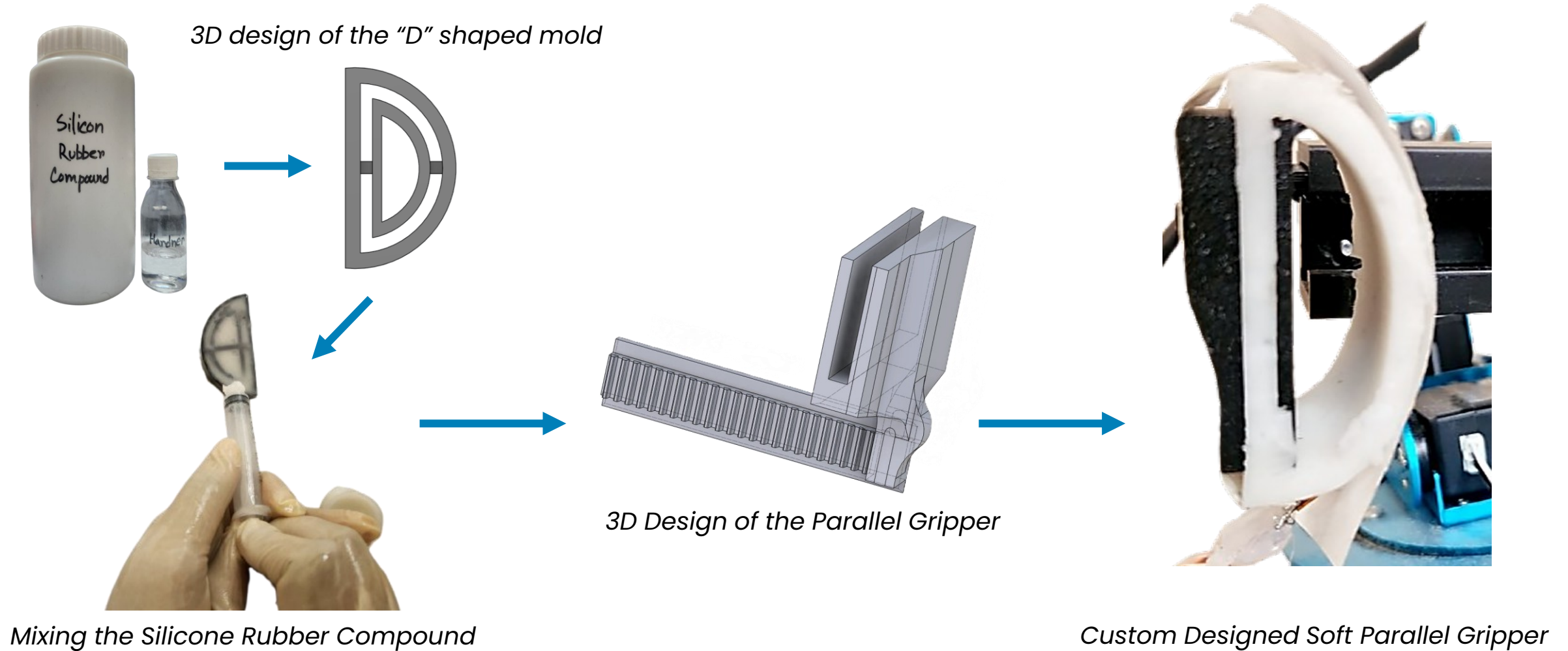
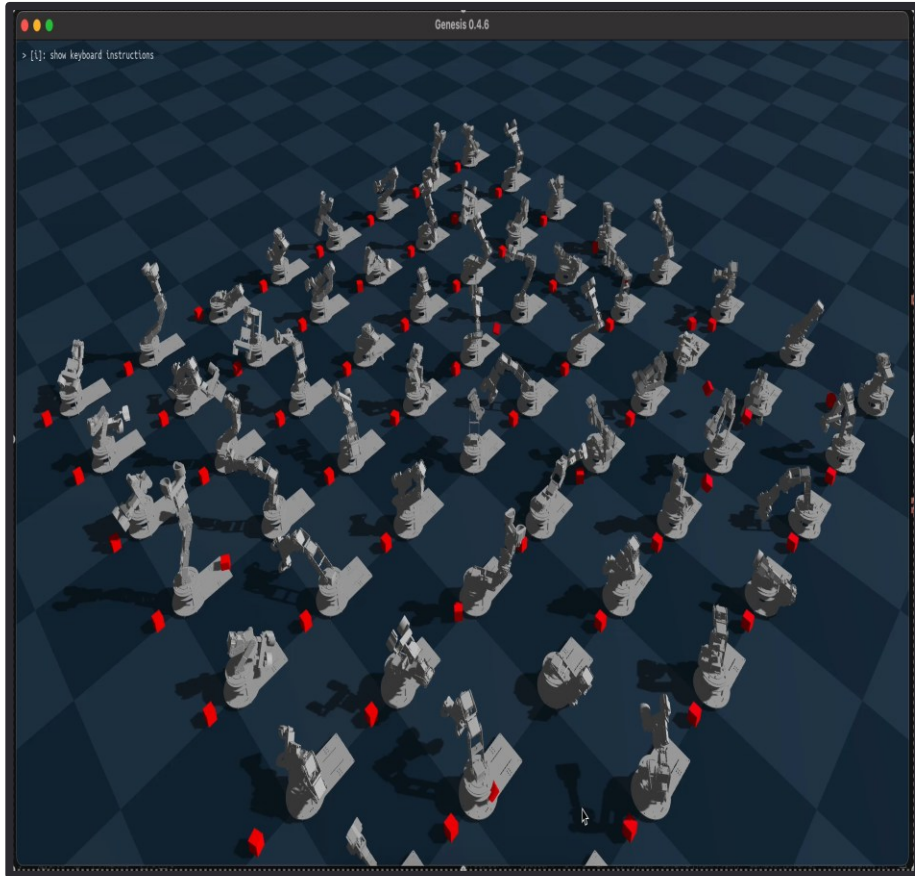
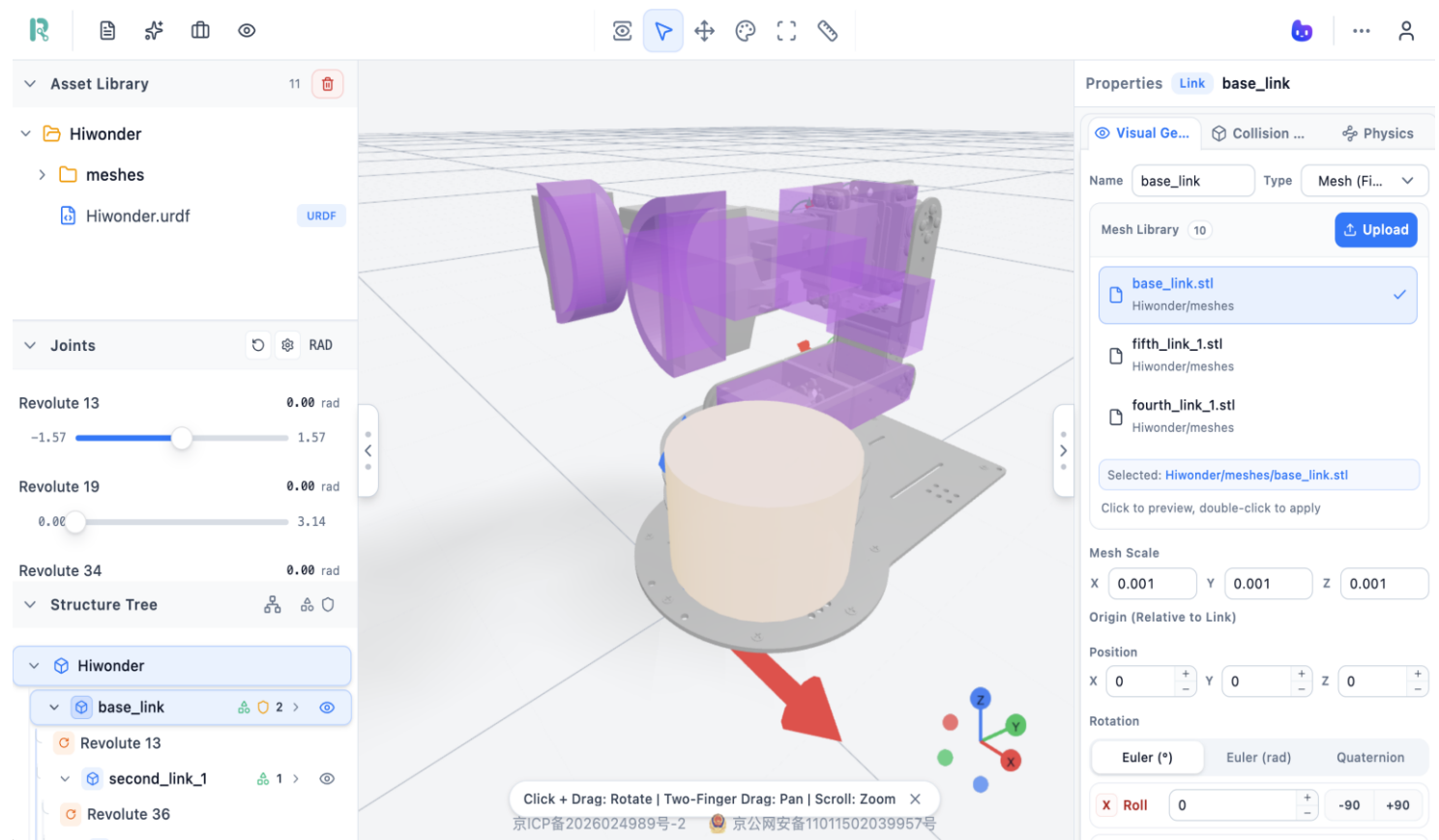


Figure 13: Soft Gripper Design and Fabrication

SOFTWARE DEVELOPMENT



(a)



(b)

Figure 14: (a) RSL_RL Dynamics Training in Genesis Sim, (b) Robot URDF Creation



IMPLEMENTATION CHALLENGES

Silicone Curing Inconsistencies

The δ_{soft} parameter was affected by **improper curing** that resulted in unstable elasticity

Dynamic Motor Angle offset

Under **different load motor** experienced different angle offsets

ADC Noise

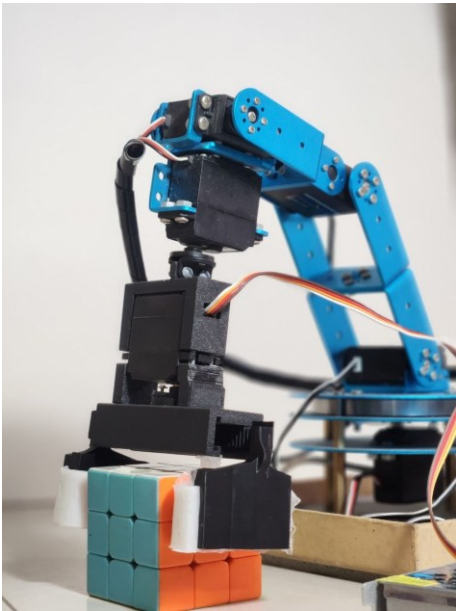
Noticeable voltages changes occurred on the module ADS1115 when the module was placed near the **high current servo motors**

Reward Shaping

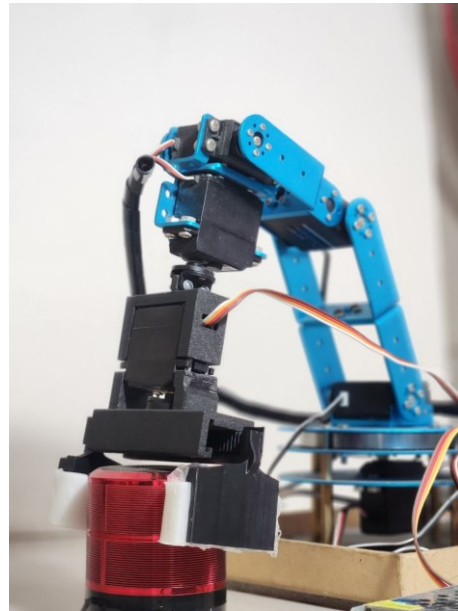
A high reward for force led to **crushing of objects** and a low reward for force led to **slippage**

EXPERIMENTAL SETUP

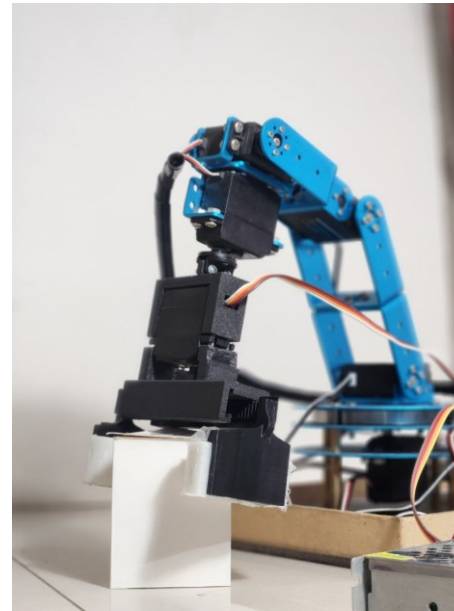
Test Objects



(a) Cube Object



(b) Cylindrical Object



(c) Triangular Object

Figure 15: Diverse Set of Test Objects

Manual Testing Interface

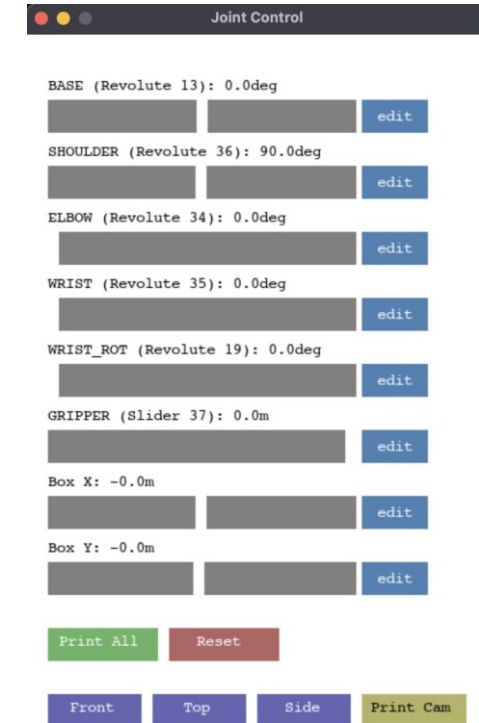
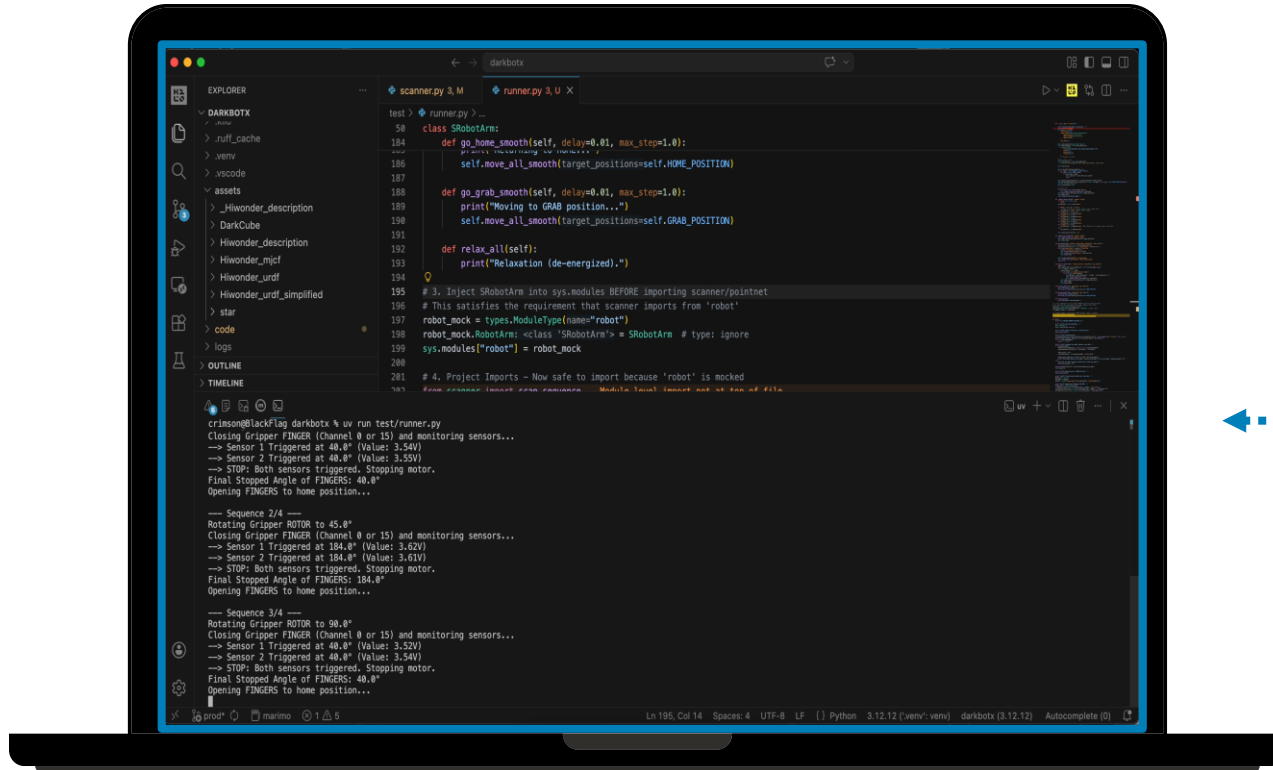
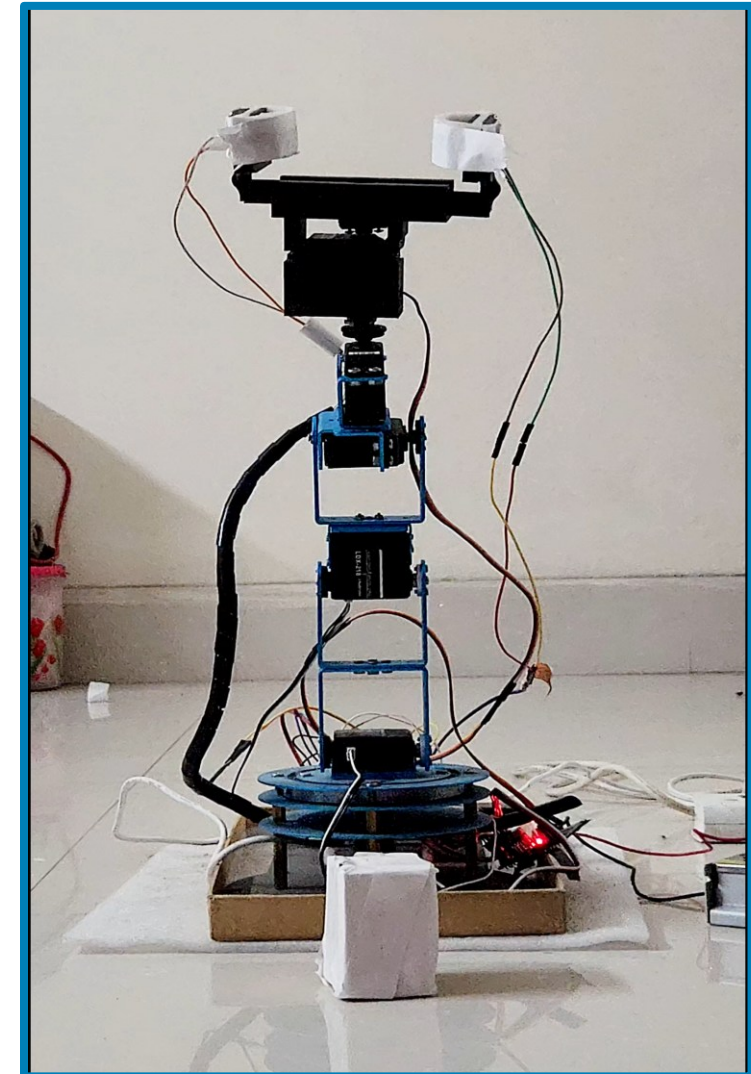


Figure 16: The 6 Motor Degree Control Interface

RESULT



SSH



RESULT

Objective 1

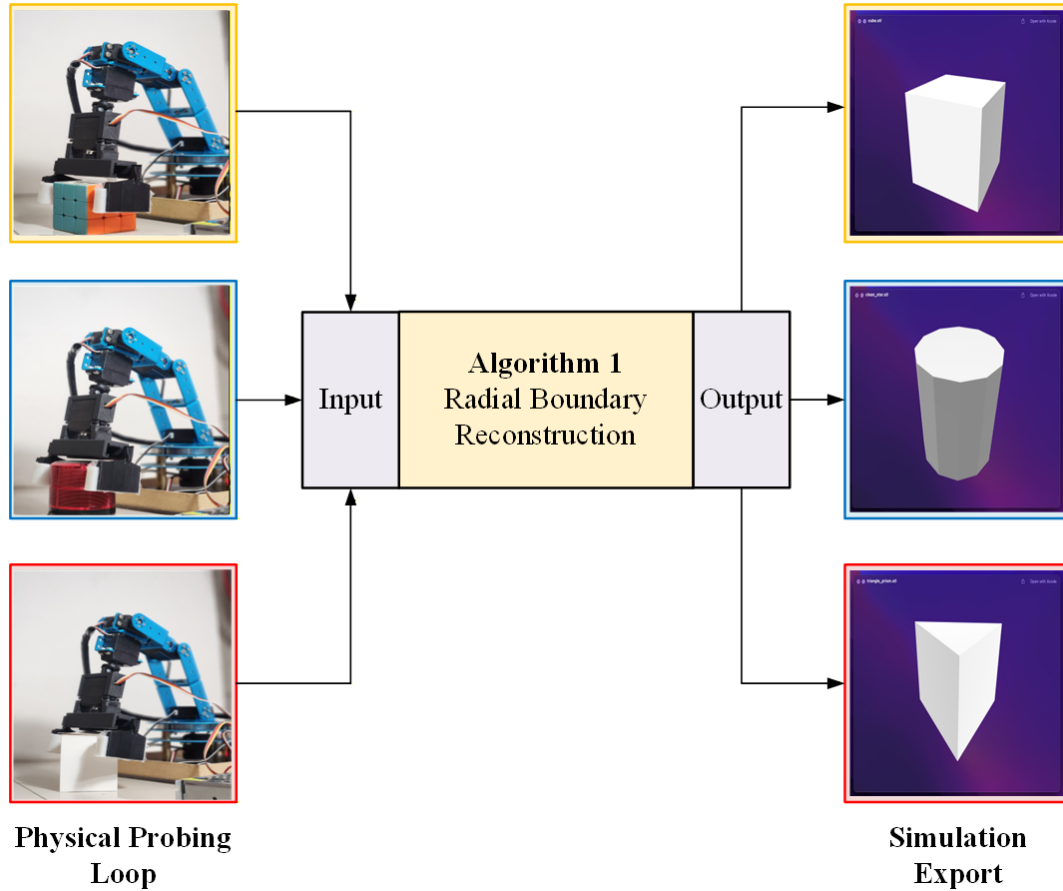


Figure 17: Comparison of the Physical Objects and Reconstructed URDF Mesh in Genesis

Objective 2

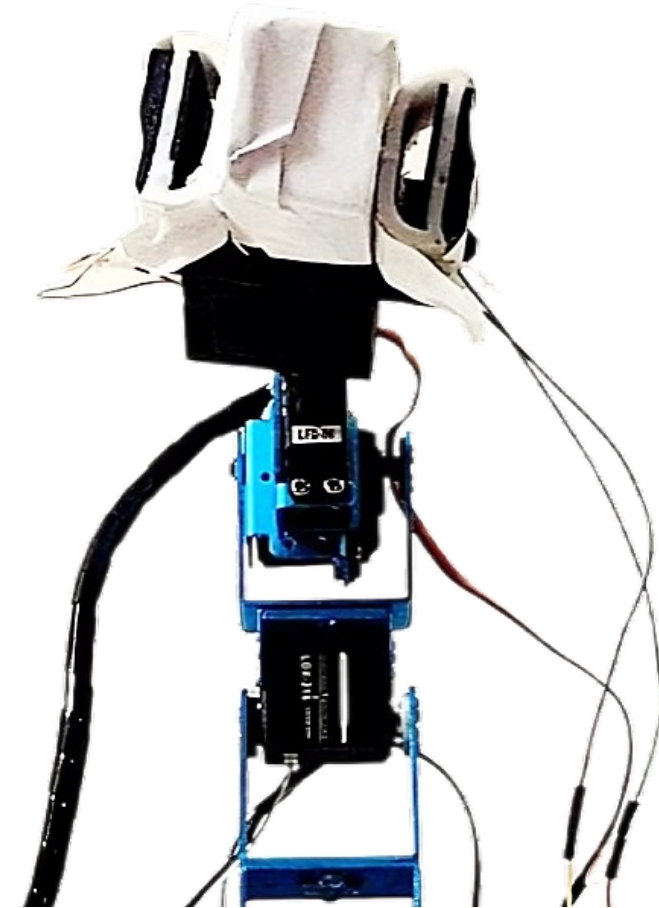


Figure 18: Optimized Visionless Grasp Execution via RSL_RL Policy

VALIDATION

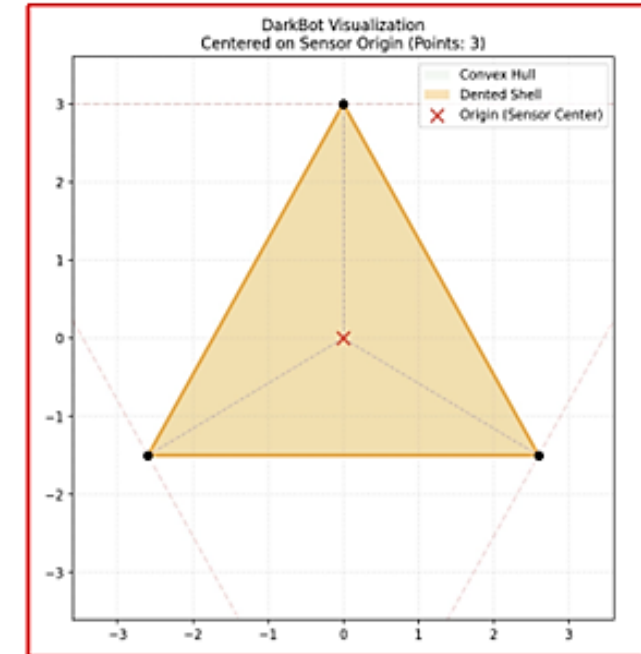
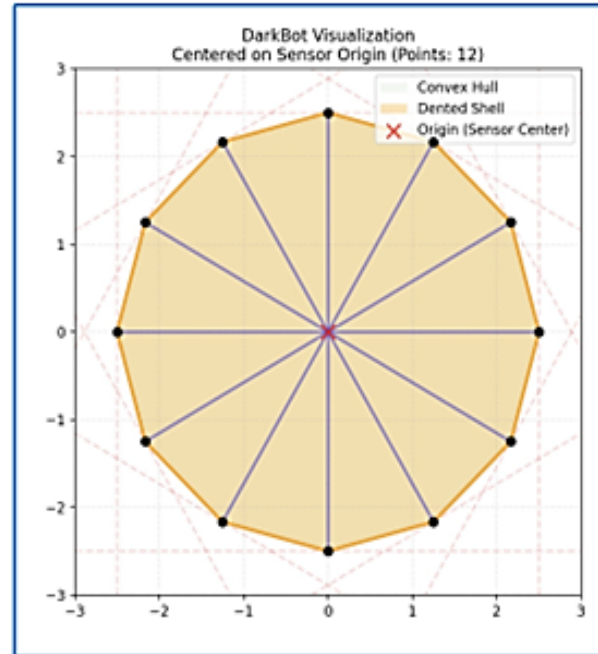
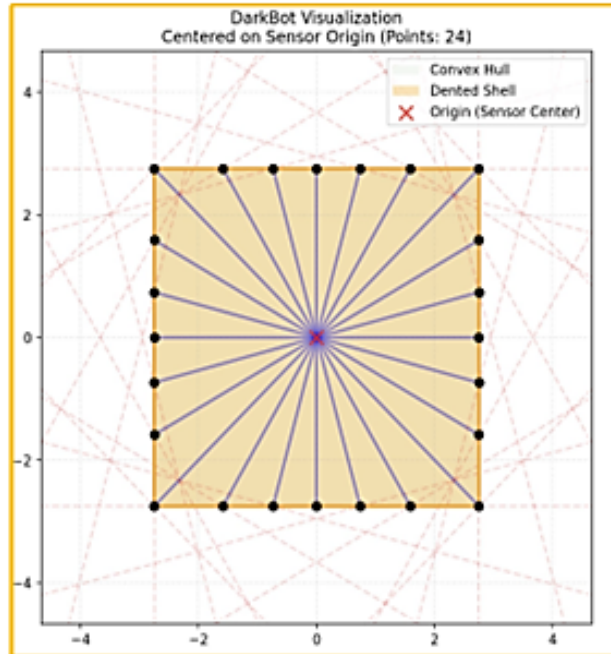


Figure 19: Visualization of The RBR Stages from Raw Radial Points to A Structured Topological Perimeter

ANALYSIS

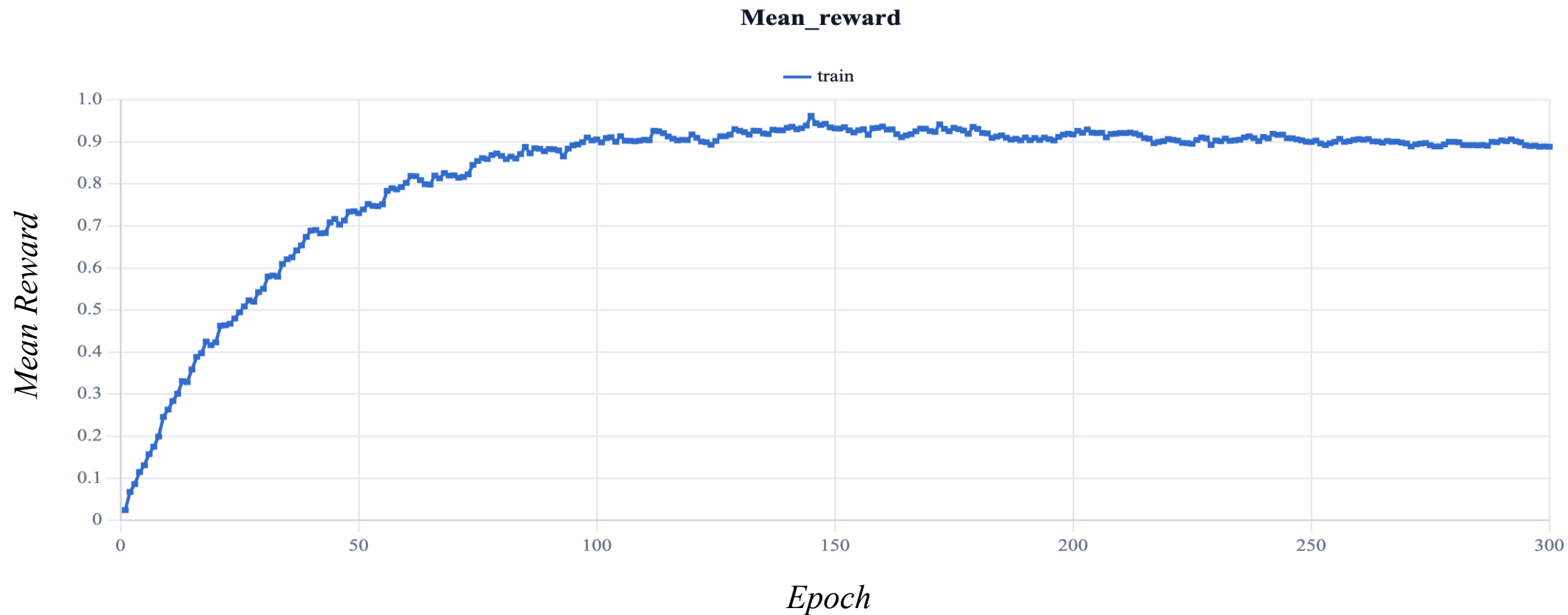


Figure 20: Policy reward convergence over a series of learning steps

ANALYSIS



Figure 21: Reduction in the loss function with the training episodes

ANALYSIS

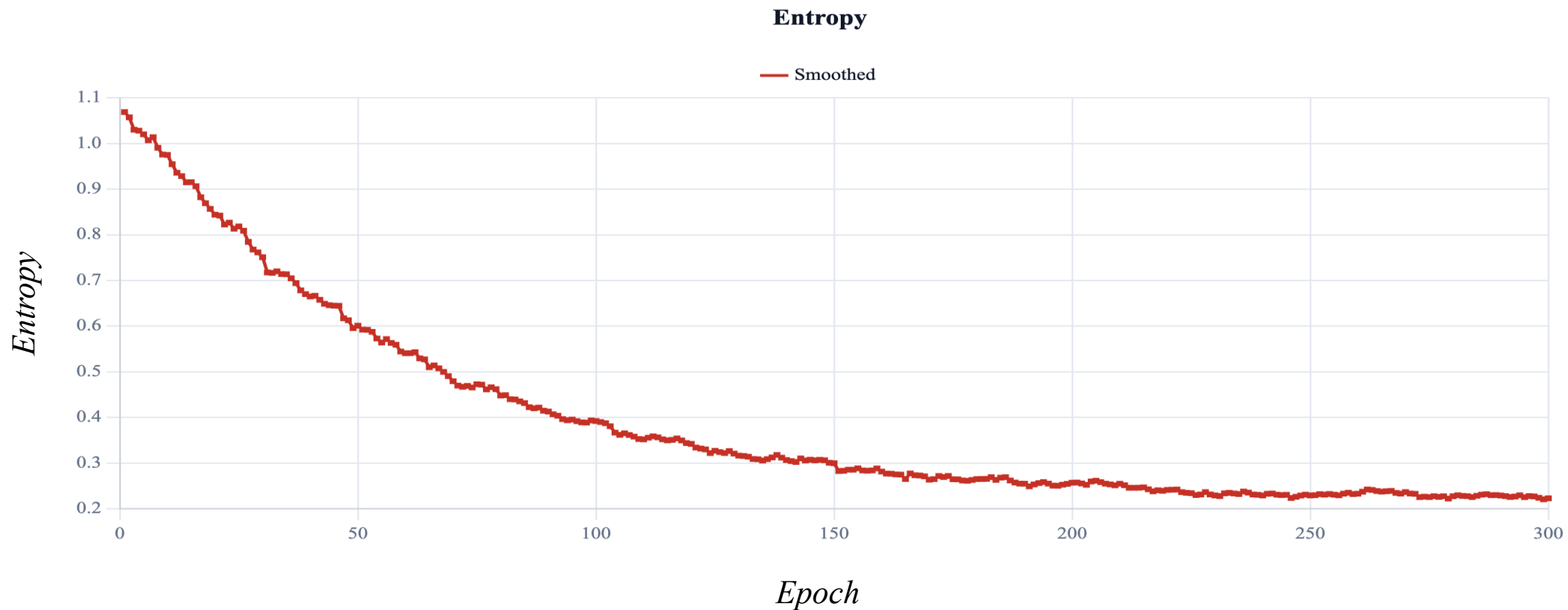
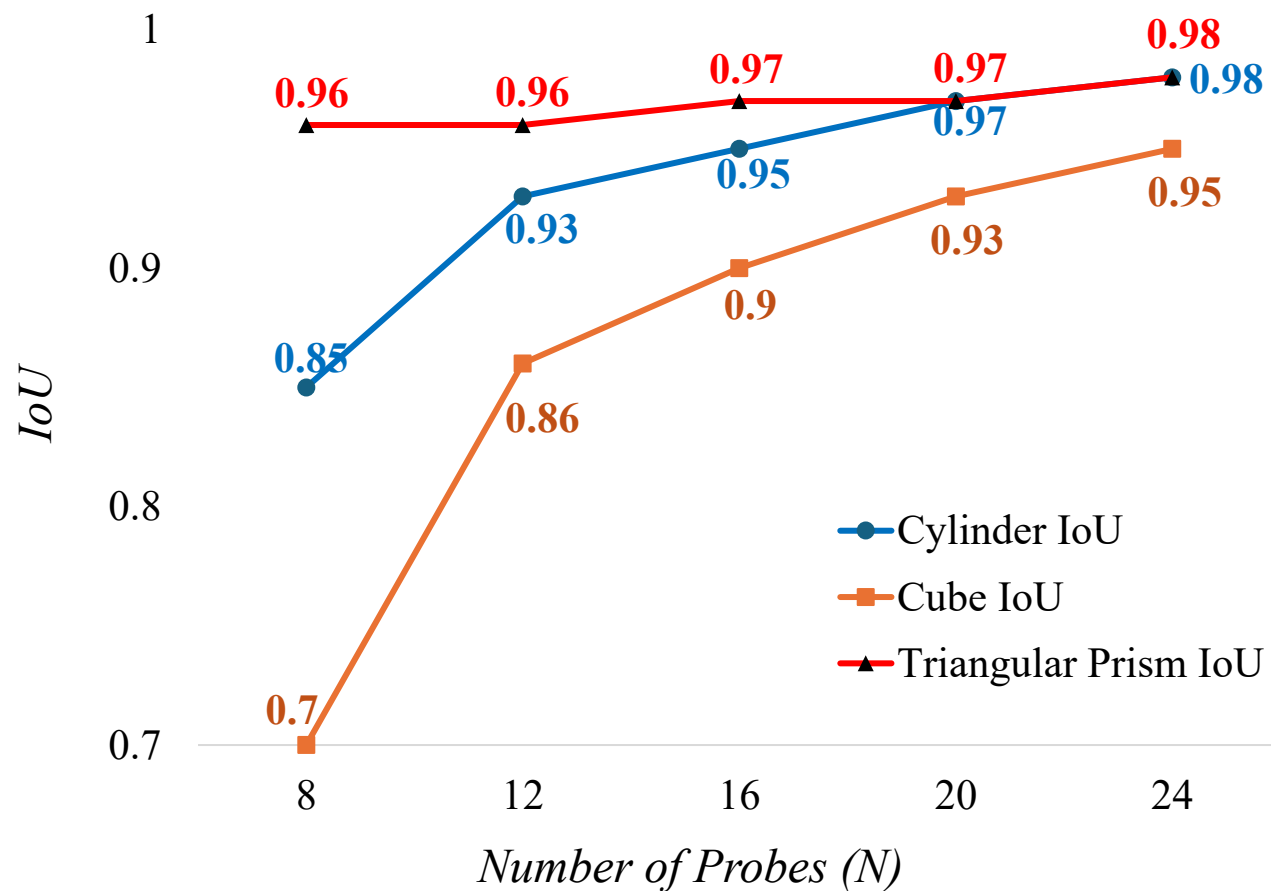


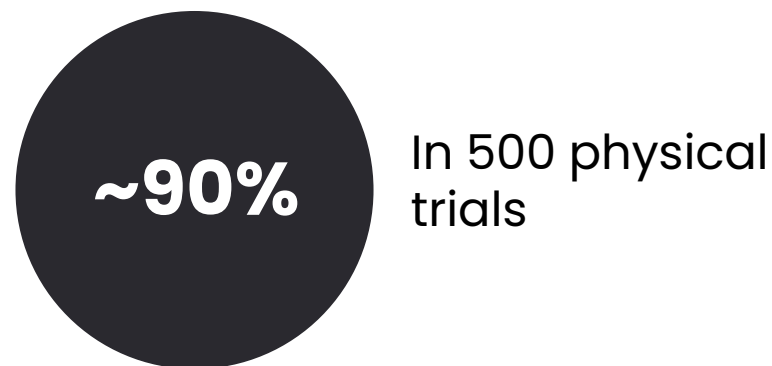
Figure 22: Plot of policy entropy decay in the training process of the RSL_RL

ANALYSIS

Convergence of IoU with Sampling Density N



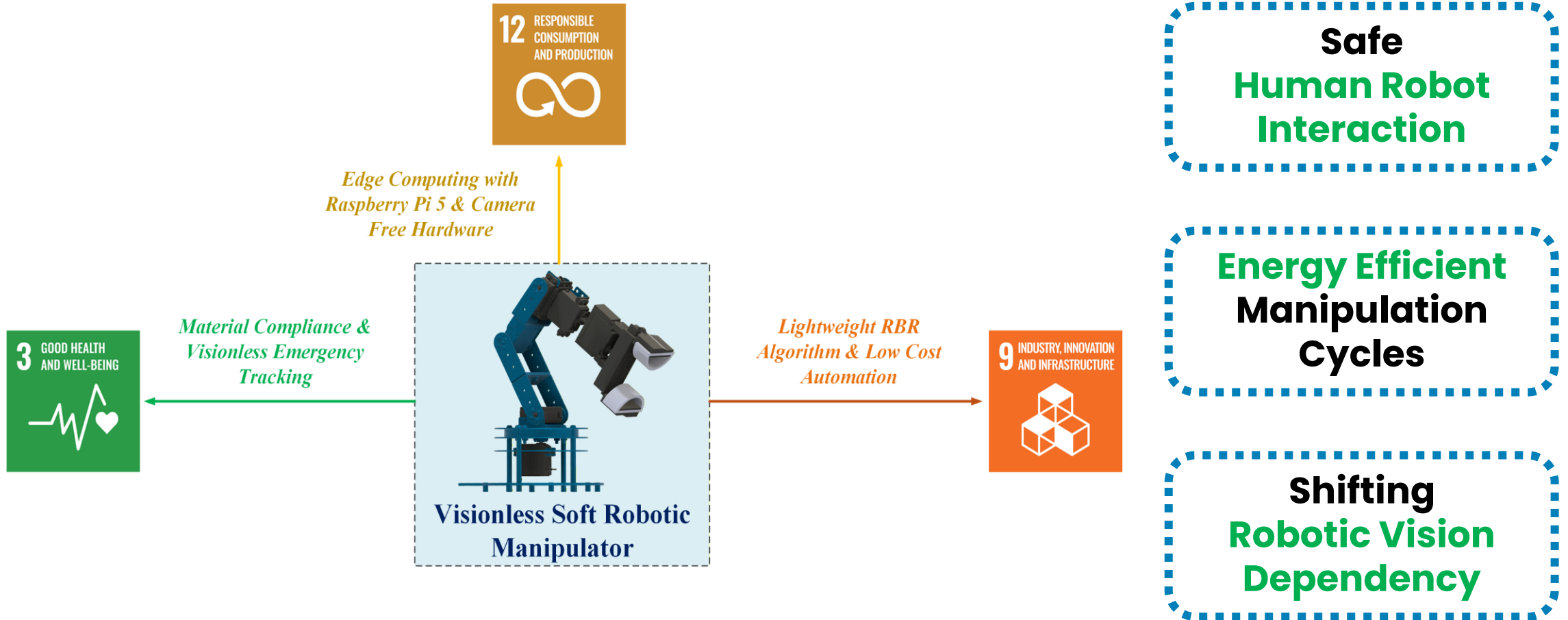
Grasp Success Rate



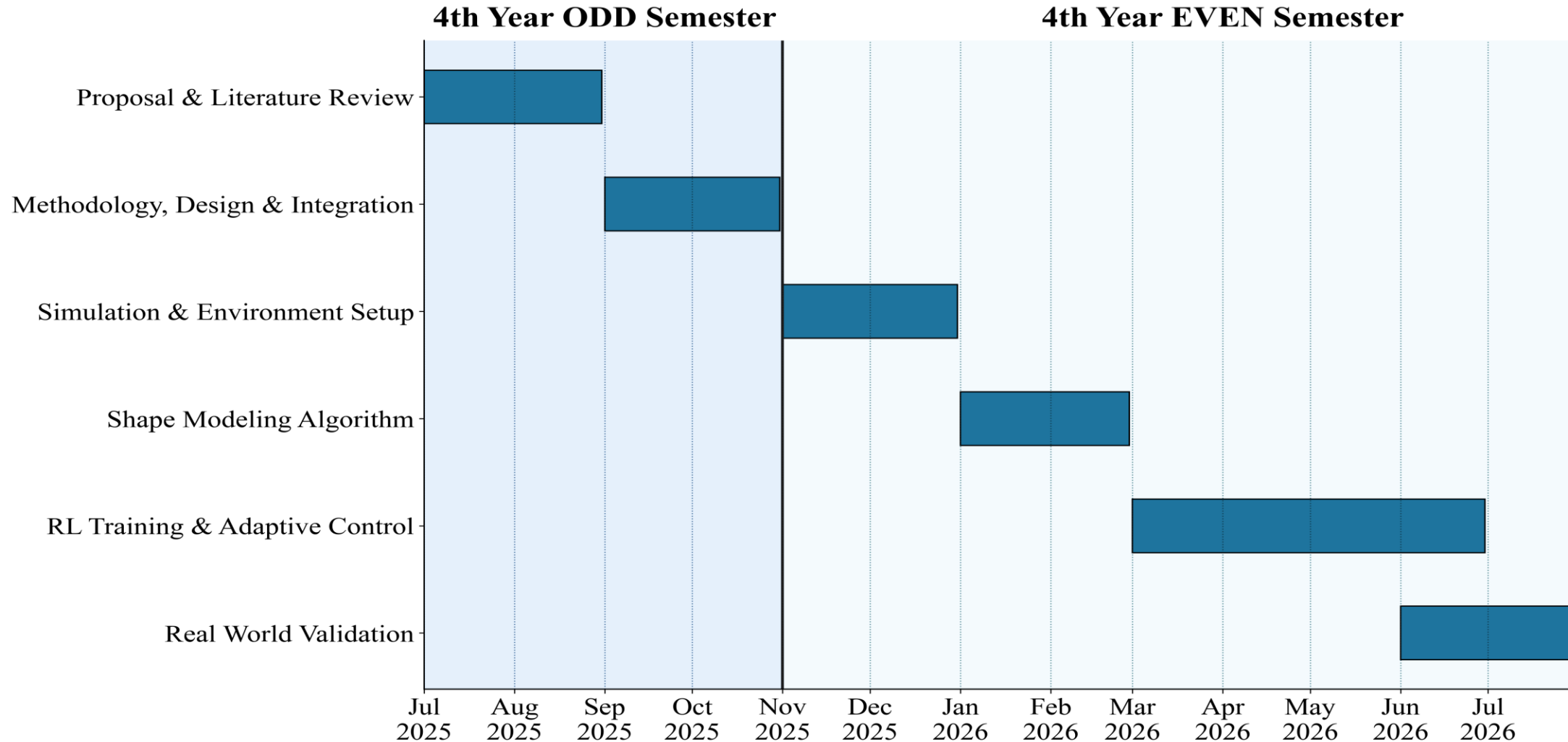
Mean Radial Error



IMPACT



PROJECT MANAGEMENT





FINANCE

SL. No.	Item	Unit Price (Including Vat and Tax)	Quantity	Total Price (Including Vat and Tax)
1.	Raspberry Pi 5 Model 16GB	BDT 28,800	1	BDT 28,800
2.	6DOF Metal Robotic Arm with Controller	BDT 27,500	1	BDT 27,500
3.	Analog Feedback Servo MG996R	BDT 1,500	2	BDT 3,000
4.	16 Channel 12-bit PWM/Servo Driver I2C Interface PCA9685	BDT 800	2	BDT 1,600
5.	Force Sensing Resistor Flexible Thin Pressure Sensor MD30-60 150g-20kg 30mm	BDT 1,200	6	BDT 7,200
6.	Flex Sensors 2.2" Original	BDT 2,250	6	BDT 13,500
7.	Silicone Rubber Compound with Hardner	BDT 7,530	2	BDT 15,060
8.	3D Printing + Accessories	BDT 6,500	1	BDT 6,500
Grand Total				BDT 1,03,160

This research was funded by Rajshahi University of Engineering & Technology.

DISCUSSION & LIMITATIONS

Successfully mimicked cephalopod

“**active touch**”

transitioning from exploratory scanning to stable manipulation
without any visual input.

But?



Assumes **homogeneous density**, CoM displacement & unstable approach for **hollow** or **non uniform objects**



Susceptibility to **corner clipping** on **sharp edged** objects



Algorithmic dependency on **fixed height extrusion** restricts the accurate reconstruction of **non prismatic** 3D structures



FUTURE WORK

Non Prismatic Reconstruction

Extending the RBR algorithm to support **variable height extrusion** for complex non prismatic 3D structures

Active Probing Policy

Integrating RL directly into the **Physical Probing Loop** to adaptively choose the next probing angle θ

Dynamic Object Handling

Training the RSL RL agent in Genesis Sim to **manage moving targets** in unstructured vision denied environments

CONCLUSION



Dark_bot

“

Established a new baseline for **vision independent soft robotics** that can **operate optimally** where traditional robotic vision fails.



THANK YOU

for Your Attention!

For further details or questions,

ABRAR FAIYAZ ANAN CHOWDHURY
2008008
Email: 2008008@student.ruet.ac.bd
Phone: 01779942933

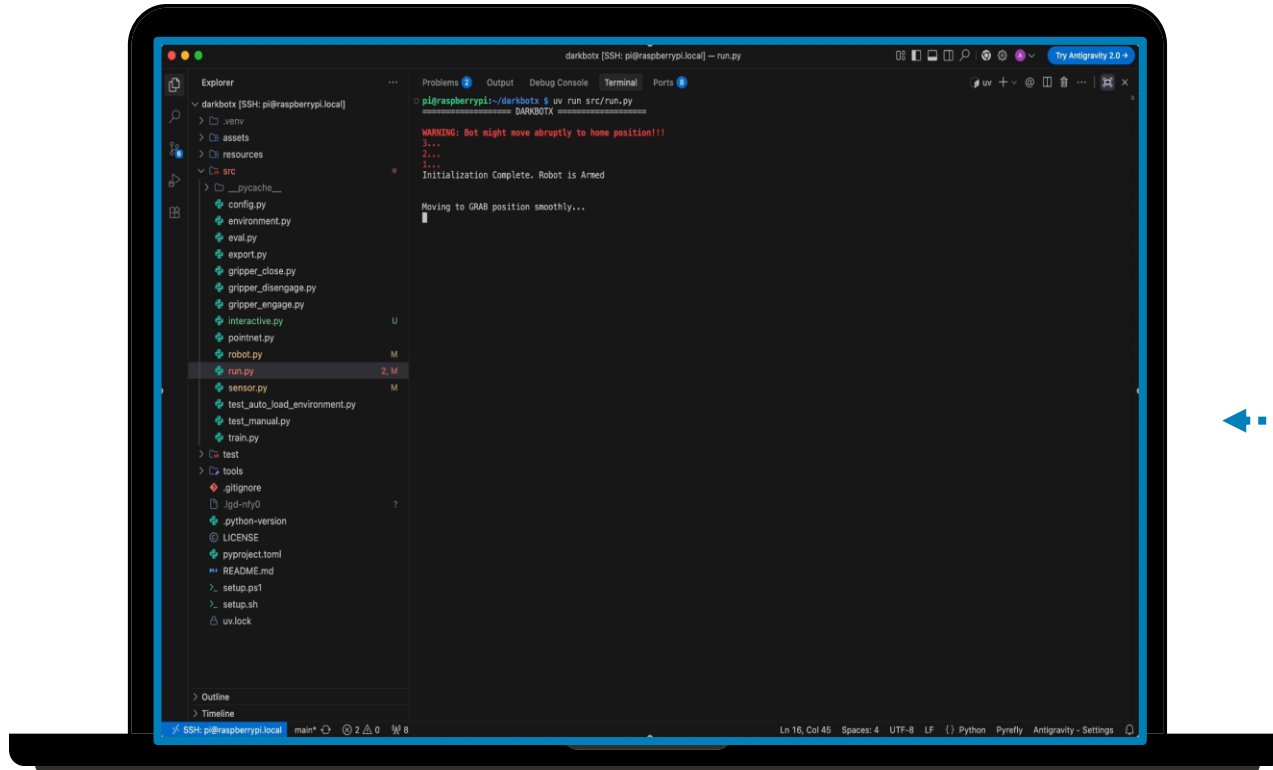
AVIK MD. EMTIAZ AREFIN
2008013
Email: 2008013@student.ruet.ac.bd
Phone: 01887818779

MD ZUNAID HOSSEN, Lecturer
Department of Mechatronics Engineering, RUET
Email: zunaid.hossen@ruet.ac.bd
Phone: 01711999250

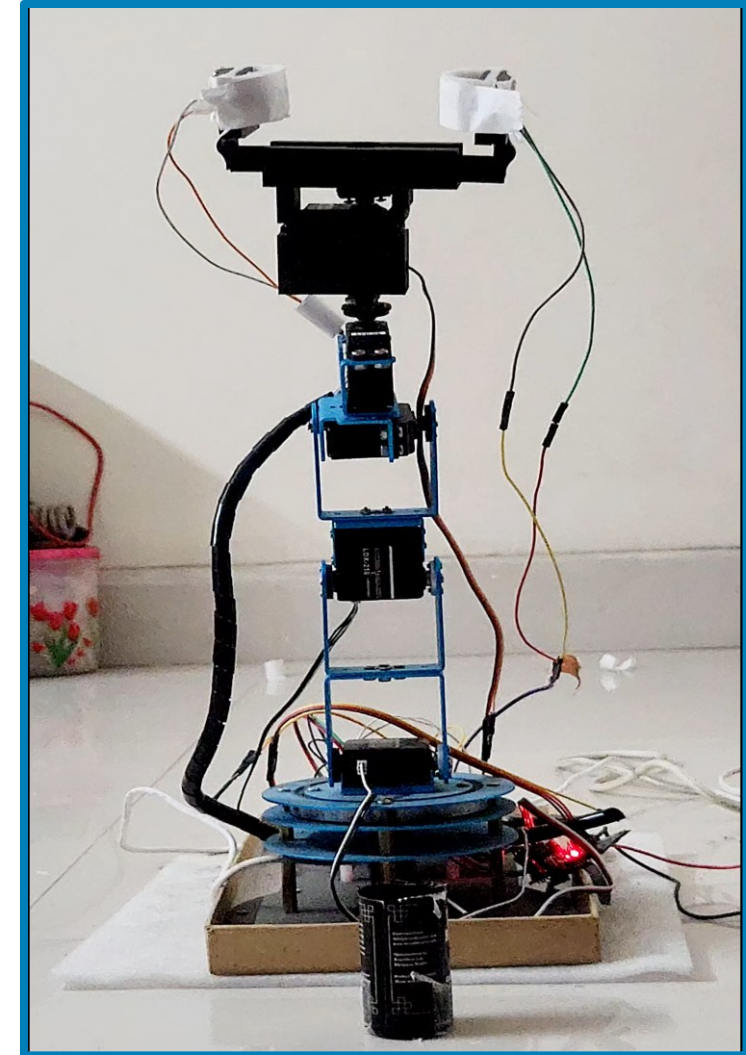


APPENDIX

RESULT



SSH





SOFTWARE COMPONENTS

Hardware Interfacing & I/O

- `adafruit-circuitpython-servokit (v1.3.22)` and `adafruit-circuitpython-ads1x15 (v3.0.2)`
- `rpi-lgpio (v0.6)`

Geometry and URDF Synthesis

- `mapbox-earcut (≥2.0.0)`, `numpy-stl (≥3.2.0)` and `urdf-parser-py`
- `trimesh`, `EarCutting`

Simulation & Machine Learning

`torch (≥2.10.0)`, `rsl-rl-lib (≥5.0.1)`, `genesis-world (≥1.2.1)`

GUI & Operator Utilities

`Pygame`, `grpcio`.



USEFUL LINKS

Repository Link:

<https://github.com/AvikArefin/darkbotx.git>

Manual Link:

<https://docs.google.com/document/d/1HCdZa46qViYCYwFxtEwEtZvLRy5nqfmjiSSAXWfae84/>

Thesis Book Draft Link:

<https://drive.google.com/drive/folders/1KYykvv3N3sAn4fL68ct7XjELVet3ldmJ?usp=sharing>